

Computer Graphics – April 9th, 2026

Name: _____

Matriculation Number: _____

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (20 pages on 10 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If needed, we staple additional blank sheets.
- No other material except a calculator is allowed!

I have read and understood the instructions above.

Erlangen, April 9th, 2026

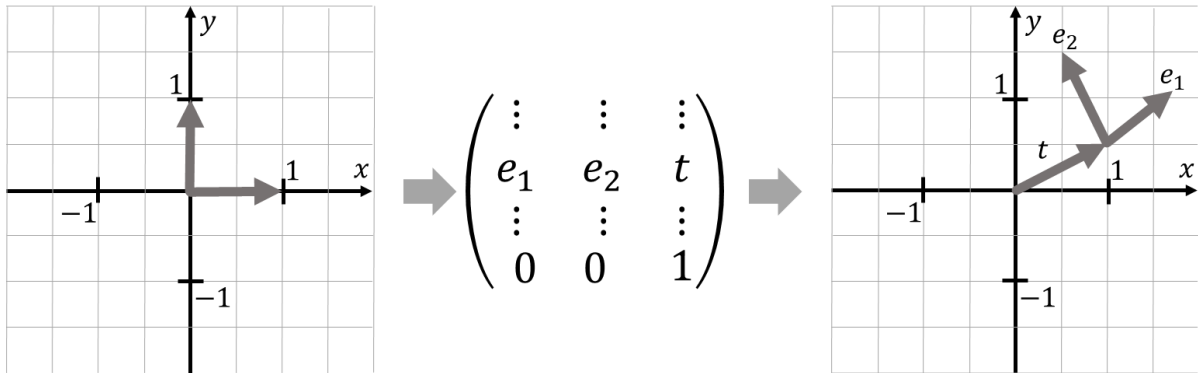
Signature

1	2	3	4	5	6	ALL
12	9	9	12	6	12	60

Grade:

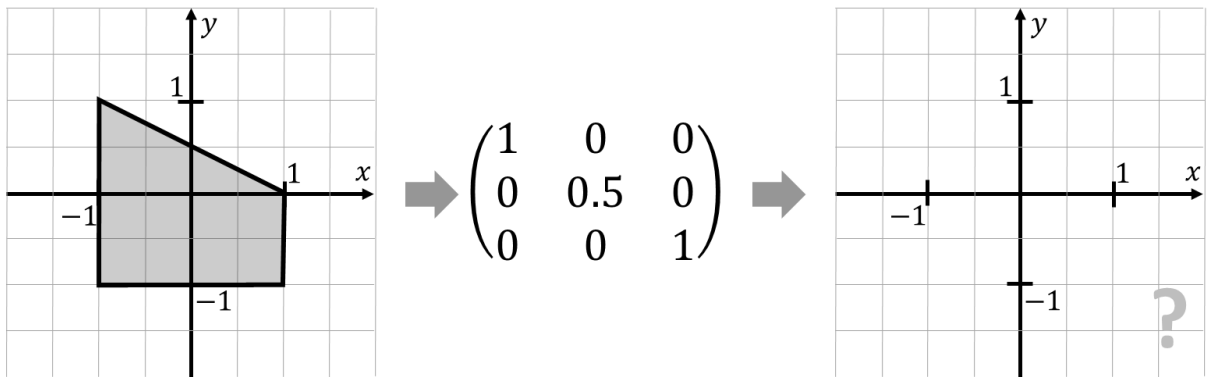
Assignment 1: Affine Mappings in 2D [12 points]

Affine mappings in **2D** can be expressed using **homogeneous coordinates** and 3×3 matrices with the following structure, where $(\vec{e}_1, \vec{e}_2)$ is the linear part (image of base vectors) and $\vec{t}$ is the translation:

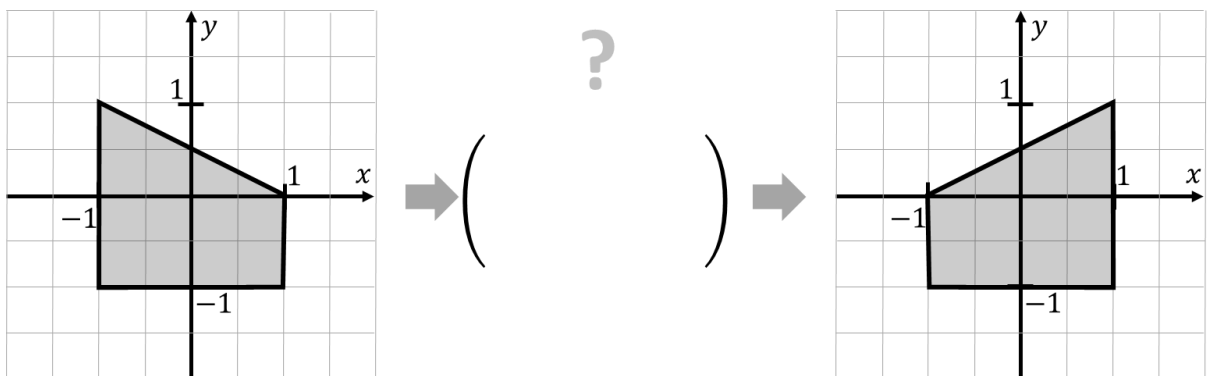


Each of the sub-assignments a) to e) deals with one mapping (in the coordinate system depicted above). It shows a shape before and after the mapping, and the corresponding matrix. Complete each of the sub-assignments by **filling in the missing part** marked with a question mark.

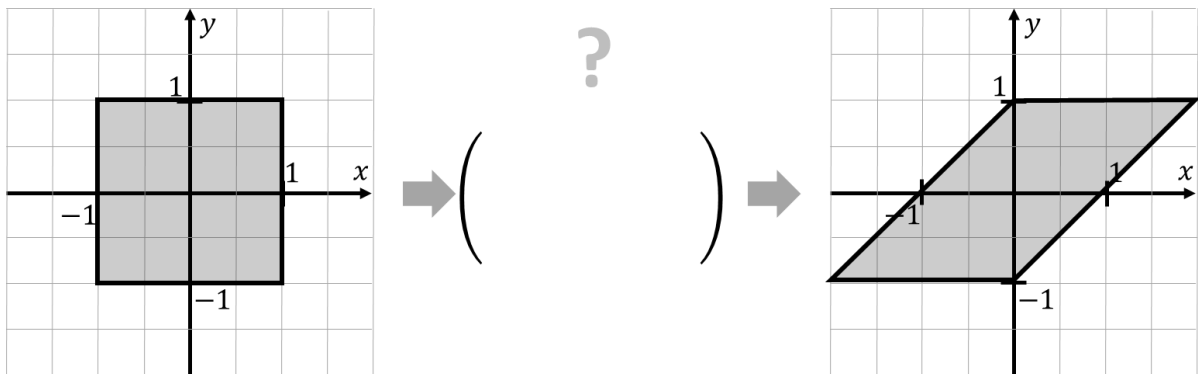
a) [1P]



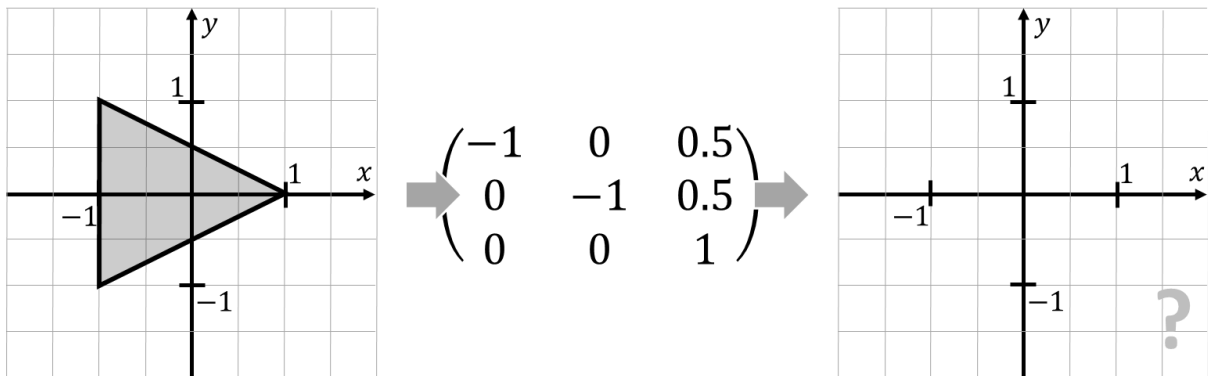
b) [1P]



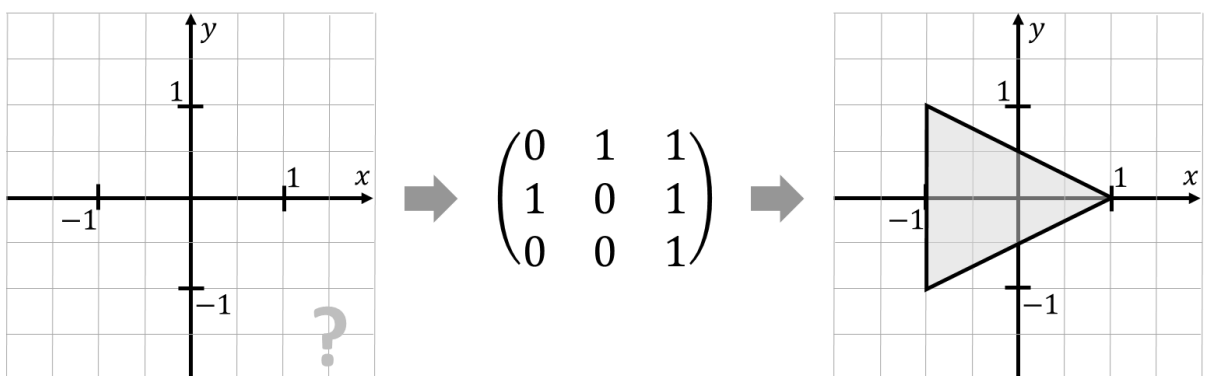
c) [2P]



d) [2P]



e) [2P]



f) **[2P]** Which properties does the mapping described by the following homogeneous matrix fulfil (multiple choices possible)?

$$\begin{pmatrix} 1 & 0.5 & 0 \\ 0 & 1 & 0.5 \\ 0 & 0 & 1 \end{pmatrix} \text{ is}$$

☐

linear

☐

rigid

☐

affine

☐

a shearing and a translation

g) **[2P]** Explain what the following product of homogeneous 2D transformation does:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

Assignment 2: Perspective [9 points]

Given are two projection matrices

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

$$Q = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

a) [1P] Do P and Q describe a point or a parallel perspective?

P : ☐ point perspective ☐ parallel perspective

Q : ☐ point perspective ☐ parallel perspective

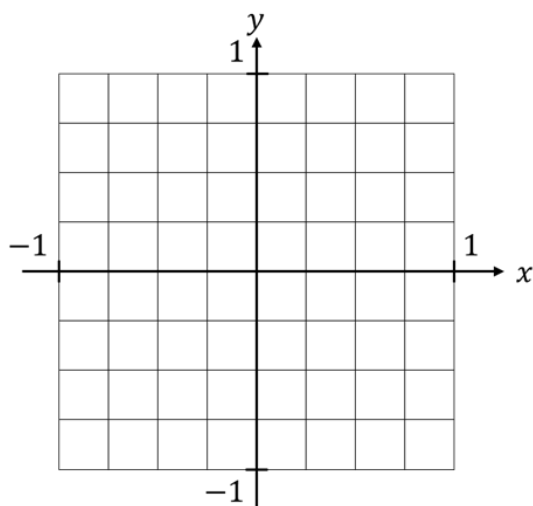
b) [6P] Given a line AB in camera coordinates with $A = (-1, -1, -1)$ and $B = (1, 1, -2)$.

Apply matrices P and Q to transform line AB into screen coordinates. Also transform the midpoint M of A and B .

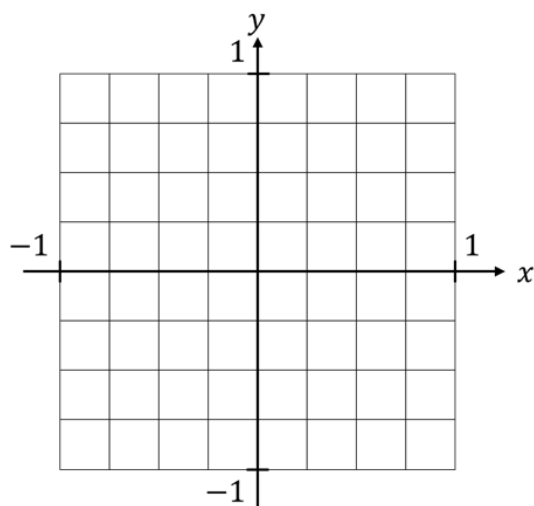
Point	Screen coordinates with Projection P	Screen coordinates with Projection Q
$A = (-1, -1, -1)$		
$B = (1, 1, -2)$		
$M = (0, 0, -1.5)$		

c) [1P] Draw the projections of the line AB , as well as M into the following grids.

Projection P



Projection Q

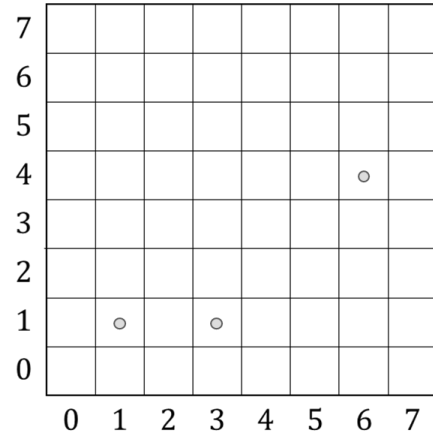


d) [1P] Look at the relative position of M with respect to A and B . Is it also the midpoint in the projection? Which fundamental difference between affine and projective mappings does this show?

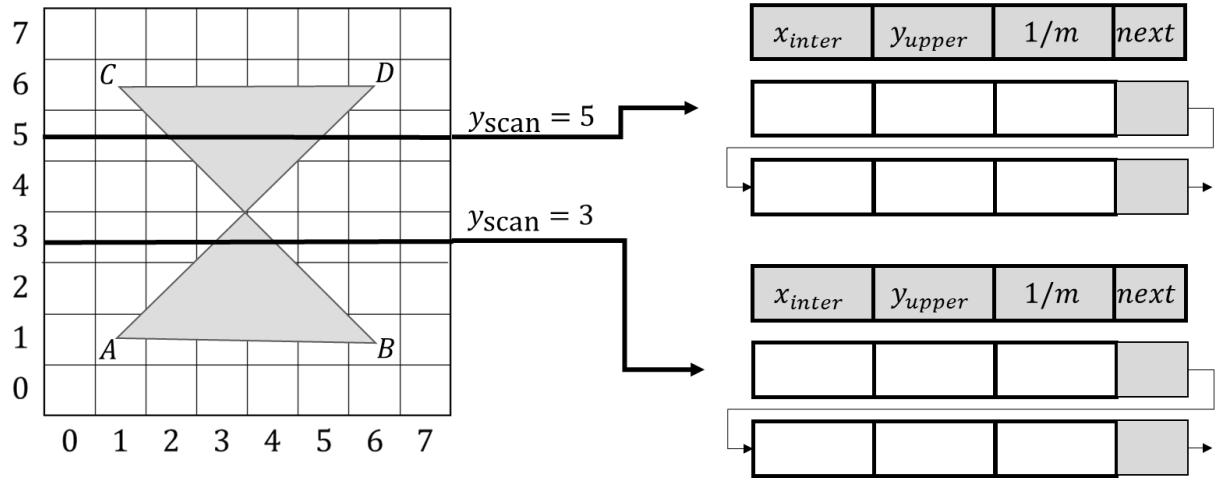
Assignment 3: Rasterization [9 points]

a) **[5P]** Given the following edge table, describing a polygon with **five** edges. Sketch the corresponding polygon in the grid on the right. For your convenience, the starting points of the edges from the edge table are already marked.

y_{lower}	x_{lower}	y_{upper}	$1/m$	$next$
1	1	6	0	
1	3	4	1	
4	6	6	0	

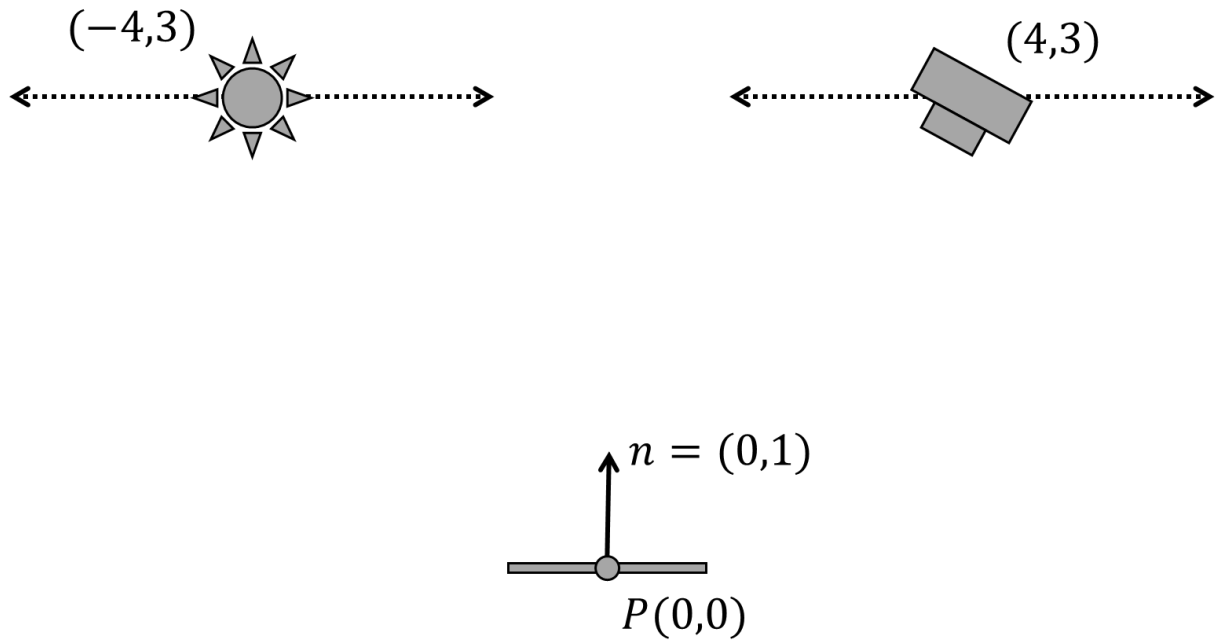


b) [4P] Given the following polygon $ABCD$, fill in the **Active Edge Table** for the scanlines $y_{\text{scan}} = 3$ and $y_{\text{scan}} = 5$.



Assignment 4: Phong Lighting & Shading [12 points]

The image below shows a point P at $(0,0)$ belonging to a surface with the normal vector $(0,1)$. The surface is lit by a point light located at $(-4,3)$ and captured by a camera located at $(4,3)$. The light coming from the point light is not attenuated. For subtasks a) and b), you can ignore the dashed arrows.



- a) **[3P]** For the point P , write down the respective *normalized* light vector l , view vector v and reflection vector r !

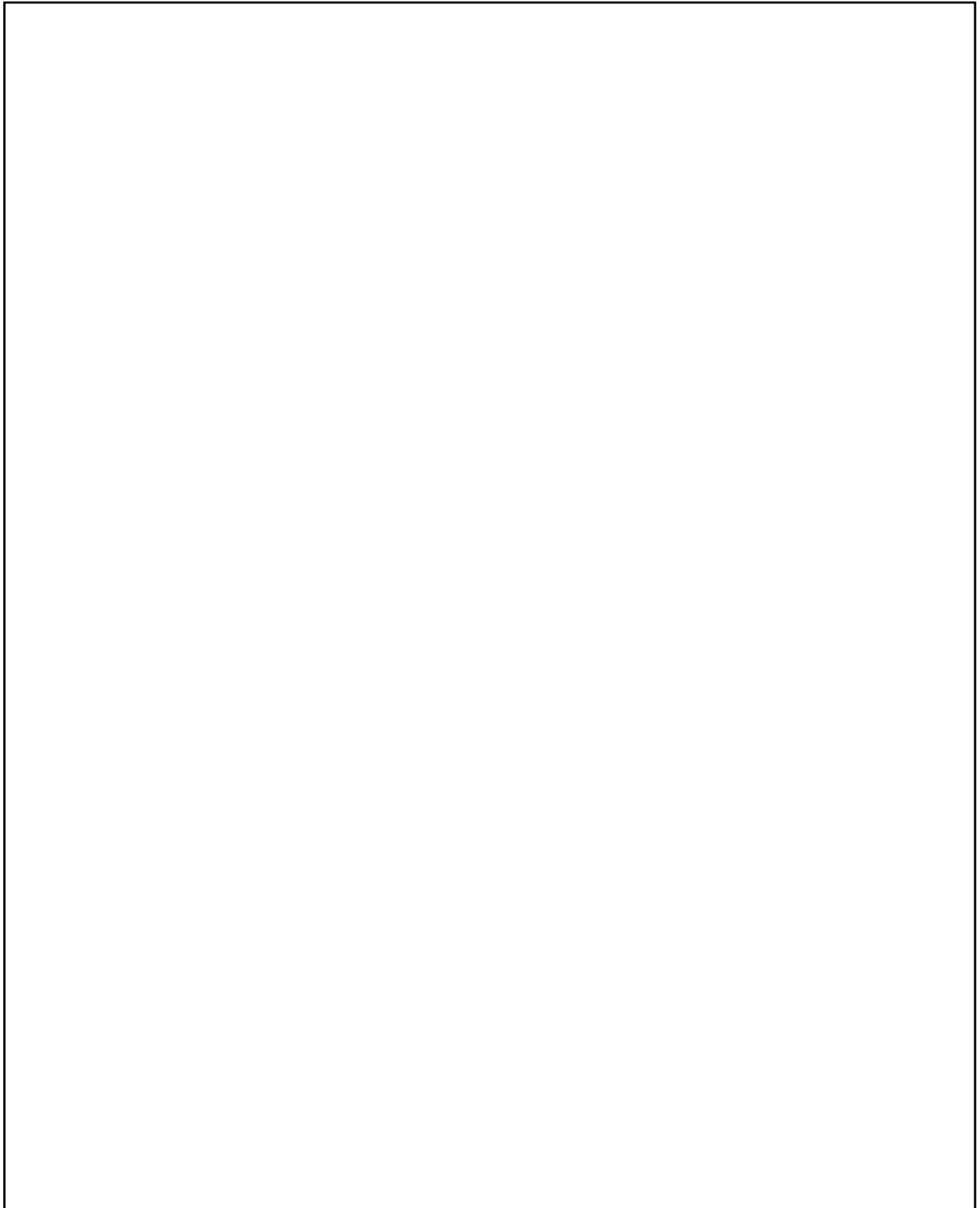
b) **[5P]** Let's assume the following Phong Lighting parameters for the surface:

$$k_{amb} = (0,0,0), k_{diff} = (0,0,1), k_{spec} = (1,0,0), n_{shiny} = 2$$

The point light has an intensity $I = (25,25,25)$ with quadratic falloff with respect to distance:

$$I_{in}(d) = I/d^2.$$

Compute the color of point P !



- c) **[4P]** Next, assume that both the point light and the camera are moved along the dashed lines. For both the diffuse and the specular part, state in the following table what happens for the four movements, starting at the initial position shown in the image.

How does the **diffuse component** change for each of the following four position changes? In each row, mark one of the possibilities!

		diffuse does not change	diffuse gets darker	diffuse gets brighter
light	moves left			
	moves right			
camera	moves left			
	moves right			

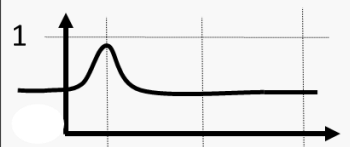
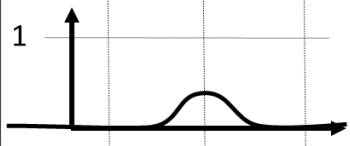
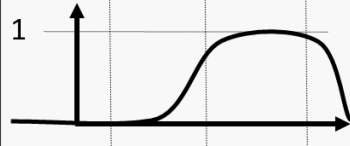
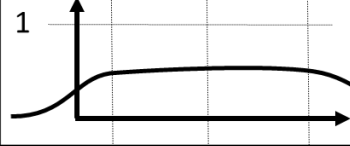
How does the **specular component** change for each of the following four position changes? In each row, mark one of the possibilities!

		specular does not change	specular gets darker	specular gets brighter
light	moves left			
	moves right			
camera	moves left			
	moves right			

Assignment 5: Colors [6 points]

a) [4P] Given the following color spectra, write down the corresponding colors and their values in RGB and HSV space (hue, saturation, value).

- For **Color name** you can choose from “white”, “grey”, “black”, “light blue”, “light cyan”, “dark magenta”, “dark green”, “yellow”, “apricot”, or “mustard”.
- **(R,G,B)** values can be 0, 0.5, or 1.0.
- For **Hue**, choose multiples of 60°, for **Saturation** and **Value** 0%, 50%, or 100%.

Spectrum	Color name	(R,G,B)	HSV-values
			
			
			
			
blue green red			

Hint: Corresponding lecture slides

Color Models

– RGB → HSV (cone model)

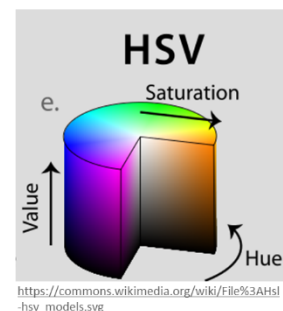
$$- V = \max(R, G, B) = \max$$

$$- S = \frac{\max - \min}{\max}$$

$$- H = \begin{cases} 60 \frac{G-B}{\max - \min} & \text{if } \max\{R, G, B\} = R \\ 60 \frac{(B-R)}{\max - \min} + 120 & \text{if } \max\{R, G, B\} = G \\ 60 \frac{(R-G)}{\max - \min} + 240 & \text{if } \max\{R, G, B\} = B \end{cases}$$

– Color pickers often use HSL, which is a variant of HSV (see definition [here](#))

$$- H \text{ and } S \text{ as for HSV, } L = \frac{\max + \min}{2}$$



0°	red
60°	yellow
120°	green
180°	cyan
240°	blue
300°	magenta

b) **[2P]** With alpha-blending, a color $C = (R, G, B, \alpha)$ is applied to a pixel with color $C_{old} = (R_{old}, G_{old}, B_{old})$ with the formula

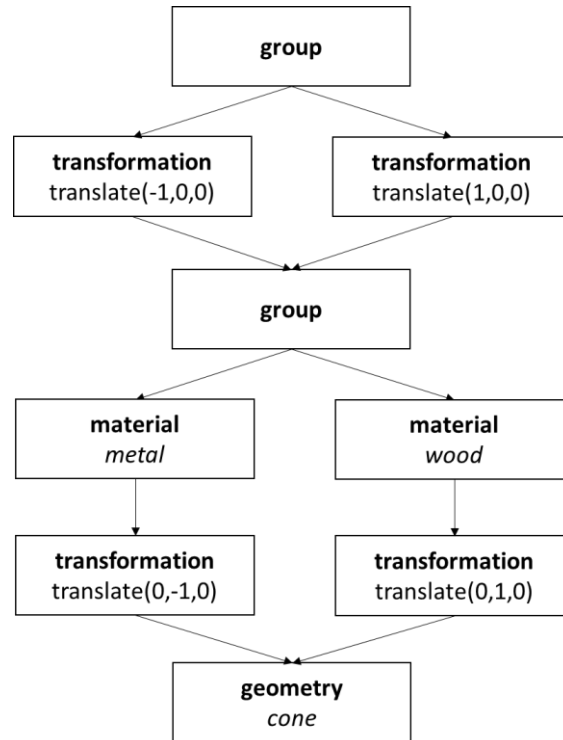
$$(R_{new}, G_{new}, B_{new}) = \alpha(R, G, B) + (1 - \alpha)(R_{old}, G_{old}, B_{old})$$

We now look at a framebuffer pixel with RGB-color (0,0,0). Use alpha-blending to apply first the RGBA-color (1,0,0,1) and then (0,1,0,0.5) and write down the resulting frame buffer colors:

Pixel value	Color to apply	Result
(0,0,0)	(1,0,0,1)	
	(0,1,0,0.5)	

Assignment 6: Scene Graphs and Rotations [12 points]

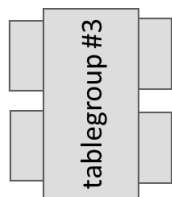
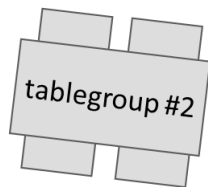
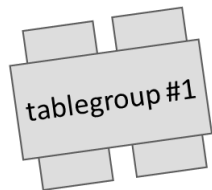
a) [4P] Given the following scene graph, describing four different instances of a cone.



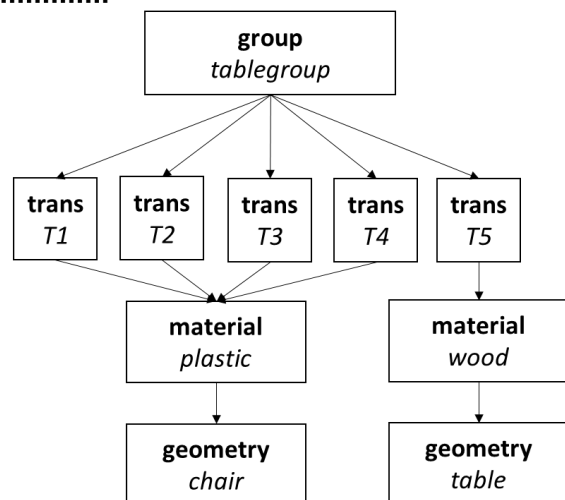
Fill in the material and transformations of the four objects

Instance	Material	Transformation	Geometry
#1			cone
#2			cone
#3			cone
#4			cone

b) [3P] Next we want to model a little restaurant scene consisting of four table groups, each with a table and four chairs. The subtree for a table group is given on the right. Extend the scene graph to describe the entire scene.



extend
scene graph
here



c) **[2P]** For scene traversal, a matrix stack is used to accumulate the transformations during depth traversal. Fill in the code for the push() method of the matrix stack (in pseudo code).

```
class MatrixStack {
    Matrix stack[];

    push(Matrix M) {
        // use this.stack.top() to get the top matrix
        // use this.stack.push(M) to push matrix M onto stack
        // use mul(A,B) to multiply matrices A and B

    }
    ...
}
```

d) **[2P]** Write down three different ways to describe a 3D rotation

1.

2.

3.

e) **[1P]** Write down a rotation representation that is well suited for interpolation

